Visual Degradation Breaks a Fine-Tuned Vision-Language Model but Not a Frozen Self-Supervised Encoder: A Robustness Study for Roadway Flood Monitoring

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***Abstract*—Roadway flood monitoring from CCTV footage is a natural application for small, edge-deployable vision-language models (VLMs), but field cameras encounter blur, low light, and sensor noise that clean benchmarks do not capture. We stress-test a QLoRA fine-tuned SmolVLM2-2.2B and a frozen I-JEPA encoder with a linear classifier, both trained for three-class flood severity classification on the same real-world benchmark, under Gaussian blur, reduced brightness, additive pixel noise, and all three applied together. Each individual perturbation degrades the fine-tuned VLM, and combining them compounds well beyond any one alone: macro-recall falls from 67.8% clean to 34.4%, and recall on the most severe flood class falls to 0.0%. A paired McNemar test confirms the combined condition is significantly worse than every individual condition, including noise alone ($p = 0.011$). The frozen encoder is essentially unchanged under noise, the one condition run for both models. Fine-tuning on clean imagery does not by itself confer robustness to compounding field conditions, and this failure stays invisible when perturbations are tested one at a time, as standard corruption benchmarks do.**

***Index Terms*—vision-language models, robustness, self-supervised learning, flood monitoring, edge deployment**

I. Introduction

CCTV cameras already installed for traffic monitoring are a low-cost sensing option for automated roadway flood detection. Small, open-weight vision-language models (VLMs) are attractive here because they run on-device, without the latency or connectivity requirements of large API-hosted models. But field cameras do not produce clean, well-lit imagery: motion blur, low light during storms, and sensor noise are normal operating conditions, not edge cases. A model that scores well on clean held-out data but fails under these conditions is unsafe for a life-safety application. Most robustness studies, including our own prior work on this task, test perturbations one at a time and never combined.

This paper tests one question: how does a fine-tuned VLM's flood severity classification degrade when blur, low light, and noise are applied simultaneously, and does a frozen self-supervised encoder degrade the same way? We fine-tune SmolVLM2-2.2B with QLoRA and pair a frozen I-JEPA encoder with a logistic regression head, both trained on the same real-world flood data, then re-evaluate both under Gaussian blur, reduced brightness, additive pixel noise, and all three combined. The fine-tuned VLM's degradation compounds sharply under the combined condition, well past what any individual perturbation predicts, including total loss of recall on the most severe flood class. The frozen encoder's performance under the one condition tested for both approaches is essentially unaffected. This compound-degradation result has not been reported for small VLMs on this task, and it shows that testing perturbations individually understates the real failure risk.

II. Related Work

Small VLMs such as SmolVLM [1] are candidates for edge-deployable visual reasoning. Prior work on this same flood classification task showed that fine-tuning on a blend of synthetic and real-world imagery closes a large synthetic-to-real gap present in zero-shot evaluation, and that I-JEPA [2], used as a frozen encoder with a linear classifier, matches or beats the fine-tuned VLM's accuracy with much tighter seed-to-seed reproducibility. That advantage is driven mostly by trainable-parameter count, since a linear probe exposes far fewer parameters to a small real-data sample than full-model fine-tuning, rather than by the self-supervised objective itself. That comparison used only clean images, and where robustness was tested, only one perturbation at a time. This paper extends it to combined, compounding degradation, a more realistic proxy for field conditions than any single perturbation alone.

Standard corruption benchmarks, most notably ImageNet-C [3], test a wide range of corruptions but apply each independently across severities rather than combining them. Robustness to one corruption in ImageNet-C does not establish how a model behaves when several degradations co-occur, which is the normal case for an outdoor camera during a storm. Self-supervised representations are not automatically more robust to corruption than supervised ones either: Chhipa et al. [4] found SSL representations lose robustness sharply at higher corruption severities. Our result is a specific data point for this task and this encoder, not a general claim about SSL robustness.

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Fig. 1. A big-flood test image under each degradation condition. The vehicle stays visible under blur, dark, and noise individually, but the combined condition removes most of the cues distinguishing flood depth.

III. Methodology

Data and models. All experiments run on STURM-FloodDepth [5], a public dataset of real-world street photos of vehicles at varying flood depths. Its five ordinal levels are collapsed to three severity labels for this task: no flood, small flood, and big flood. Dr. Chuprov's lab at the University of Texas Rio Grande Valley (UTRGV)[1] supervised this project as part of an ongoing effort applying vision models to flood-depth estimation. Two classifiers are compared, both trained on an identical 30% real-data split: a QLoRA-adapted [6] SmolVLM2-2.2B [1], where rank-8 LoRA [7] adapters sit on the language model's attention layers while the vision encoder stays frozen, and a logistic-regression head trained on top of I-JEPA's [2] frozen ViT-H/14 patch embeddings, mean-pooled, with no gradient ever touching a language model.

Degradation protocol. Both trained classifiers are scored a second time on the exact same held-out images after four transformations: Gaussian blur (radius 3), a brightness cut to 40% of the original, additive Gaussian noise ($\sigma = 25$), and all three stacked together in that order. Figure 1 shows a big-flood test image under each condition. The submerged vehicle remains identifiable under every individual perturbation but is close to unreadable under the combined condition, which is the qualitative counterpart to the quantitative collapse reported below.

Parameters stay identical whether a perturbation is applied alone or as part of the stack, so compounding is the only thing that changes between the individual and combined conditions. The frozen encoder was only run through the noise condition because of compute limits; we did not guess at the other three.

Significance testing. Since every condition scores the same 238 images, correctness on any two conditions pairs up naturally. We run McNemar's exact test on the combined condition against clean and against each individual perturbation, using the paired 2×2 correct/incorrect table; McNemar suits paired binary outcomes and carries no normality assumption.

IV. Results

Table I reports SmolVLM2-2.2B's accuracy and macro-recall across the full four-condition degradation battery, and Figure 2 breaks the same runs down by class.

Every individual perturbation cuts SmolVLM's macro-recall, and the combined condition compounds past all of them. Blur, reduced brightness, and noise each cut macro-recall below the 67.8% clean baseline; noise alone causes the largest drop, to 41.5%. The combined condition drops further, to 34.4%, below every individual perturbation. This is not additive. Figure 2 shows why: under every degraded condition the model reverts to over-predicting "no flood" (recall at or above 94% for that class in all four degraded conditions), while small- and big-flood recall erode steadily. Under combined degradation, big-flood recall hits 0.0%: every genuinely severe flood image in the test set was misclassified. The failure is directional, not a uniform loss of accuracy, and it runs opposite to what a flood-warning system needs.

TABLE I
SmolVLM2-2.2B Accuracy / Macro-Recall Across Conditions

Condition	Accuracy	Macro-Recall
Clean	64.7%	67.8%
Blur	50.0%	54.2%
Dark	57.1%	64.1%
Noise	33.6%	41.5%
Combined	26.9%	34.4%

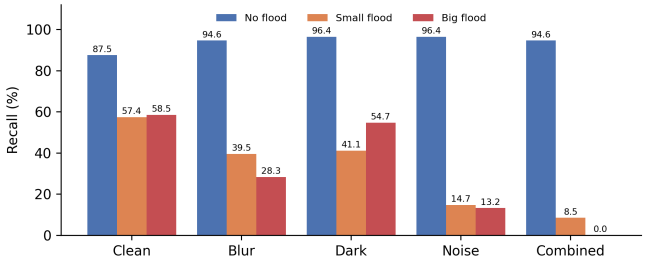


Fig. 2. SmolVLM2-2.2B per-class recall by condition. No-flood recall stays at or above 94% under every degradation while small- and big-flood recall collapse, ending at 8.5% and 0.0% under combined degradation. The model does not degrade uniformly; it retreats to a single class.

The frozen encoder is essentially unaffected by the one condition run for both models. We deliberately limited the cross-method comparison to clean and noise, the only two conditions where both classifiers were evaluated under matched settings; Table II reports both directly rather than mixing tested and untested cells in one grid. Under noise, I-JEPA's macro-recall is 58.7%, slightly above its 54.8% clean baseline. SmolVLM drops from 67.8% to 41.5% over the same transition. I-JEPA's per-class recall does not show SmolVLM's no-flood-collapse pattern: no-flood recall is unchanged (39.3% under both conditions), while small- and big-flood recall both improve slightly. A frozen representation pretrained at scale on natural images, never fine-tuned on our small clean training set, is less sensitive to the pixel statistics of that training set than a model whose entire adaptation process ran on clean images only.

The combined-condition drop is statistically significant, not sampling noise. Table III reports paired McNemar tests for SmolVLM. Every comparison against the combined condition is significant, including noise alone ($p = 0.011$), the

[1]FL Research Lab: https://angelperedo97.github.io/FL-Research-Lab/

TABLE II
MATCHED CLEAN-VS.-NOISE COMPARISON (MACRO-RECALL)

Method	Clean	Noise
SmolVLM2-2.2B (fine-tuned)	67.8%	41.5%
I-JEPA + LogReg (frozen)	54.8%	58.7%

TABLE III
PAIRED MCNEMAR TESTS, SMOLVLM2-2.2B (238 IMAGES)

Comparison	Discordant pairs	p-value
Clean vs. Combined	97 / 7	< 0.0001
Blur vs. Combined	65 / 10	< 0.0001
Dark vs. Combined	78 / 6	< 0.0001
Noise vs. Combined	26 / 10	0.011

closest individual condition by mean performance. The compounding effect is real, not an artifact of a single evaluation run on a 238-image set.

V. DISCUSSION

These results extend a clean-condition finding, that a frozen self-supervised encoder can match or beat a fine-tuned small VLM, to a robustness setting where the stakes are higher. Fine-tuning improves clean-condition accuracy but does not by itself confer robustness to compounding degradation. A model selected on clean validation data alone can pass evaluation while failing exactly during the storms and low light that motivate the application. The 0.0% big-flood recall under combined degradation is the sharpest version of this problem: it is the class a deployed system most needs to catch, and the class that fails completely once degradation compounds.

Extrapolating from the individual results would not predict the combined result. Blur (54.2%), dark (64.1%), and noise (41.5%) do not add up to 34.4%, and none of them predicts a complete collapse of big-flood recall. Robustness testing that only applies perturbations one at a time, including our own prior work on this task for the frozen encoder, understates real-world failure risk. Running the frozen encoder through the full degradation battery is the direct next step to confirm whether its noise-robustness holds under compounding.

VI. LIMITATIONS

I-JEPA was only tested under noise. Blur, dark, and combined conditions were not run for it due to compute constraints, so we cannot claim it is robust to the same combined degradation that collapses SmolVLM, only that it is robust to one of the three components alone. Both methods were evaluated at a single 30% real-data blend ratio, not the fuller range used for clean-condition results elsewhere in this line of work, and 30% is not the peak blend ratio for either method on clean data. The collapse is not an artifact of stress-testing a weak checkpoint, however: the fine-tuned instance evaluated here starts from 67.8% clean macro-recall, above the 64.5% three-seed mean of the 80% blend configuration that maximizes clean performance for this model. Repeating the battery at 80% would still be the stronger experiment. Perturbation parameters (blur radius, brightness factor, noise variance) were fixed, not swept, so the exact severity at which each method breaks remains unknown. Each condition uses one trained model instance rather than seed-averaged results. This bounds what the absolute numbers mean but not the comparisons: every condition scores the same trained model on the same 238 images, so training seed is a common offset across conditions rather than a confound between them, and McNemar's test operates on paired per-image outcomes with training held fixed. The effect sizes also sit well outside the known seed noise. Prior work on this task reports a 7.5-point macro-recall standard deviation across three seeds at the same 30% blend configuration used here, against a 33.4-point clean-to-combined drop, a factor of 4.4. The narrower noise-to-combined contrast (7.1 points) is comparable to that seed spread in magnitude, which is precisely why we test it with a paired test rather than resting on the means; it remains significant at $p = 0.011$. Seed-averaging every condition would still tighten the absolute estimates and is worth doing.

VII. CONCLUSION

We stress-tested a fine-tuned SmolVLM2-2.2B and a frozen I-JEPA encoder for roadway flood severity classification under blur, reduced brightness, additive noise, and all three combined. The fine-tuned VLM degrades under every individual condition and compounds further under combined degradation, with big-flood recall falling to 0.0%, a drop confirmed statistically significant by paired McNemar tests. The frozen encoder's performance under the one condition tested for both methods is essentially unchanged from clean. Fine-tuning on clean imagery does not by itself confer robustness to compounding field conditions, and combined-perturbation testing surfaces failures that individual testing misses, a gap standard corruption benchmarks do not address [3]. Running the frozen encoder through the full degradation battery, and testing whether self-supervised representations reliably beat fine-tuned VLMs under compounding degradation more broadly, since that advantage is not guaranteed a priori [4], are the next steps.

AI DISCLOSURE

The authors used Claude (Anthropic) as an AI writing assistant for drafting and editing portions of this manuscript.

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